

Xiaoyu Wu

PhD, MRSC | Research Fellow | Computational Chemistry and AI for Reticular Chemistry

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Academic Profile

Computational chemist developing molecular simulation, high-throughput screening, and machine-learning approaches for the digital discovery of nanoporous materials. Research spans metal-organic and covalent organic frameworks, porous liquids, adsorption and separation, framework stability and flexibility, and chemically informed AI. Experienced in interdisciplinary collaboration, student mentoring, scientific communication, and large-scale computational research.

Appointments

Research Fellow | National University of Singapore | **2023-present**

Department of Chemical and Biomolecular Engineering; supervised by Prof. Jianwen Jiang.

Education

PhD in Chemistry | University of Liverpool, United Kingdom | **2023**

Supervised by Prof. Andrew I. Cooper and Prof. Lifeng Ding.

BSc in Chemistry | Nanjing Tech University, China | **2018**

Research Focus

- AI for reticular chemistry: forward and inverse learning, active learning, deep learning, large language models, and machine-learning potentials.
- Molecular and multiscale simulation: adsorption, separation, transport, flexibility, stability, and structure-property relationships in porous materials.
- Sustainable applications: carbon capture, gas upgrading, hydrocarbon and isotope separations, sensing, and radionuclide sequestration.

Manuscripts Under Review

Linker Functionalization in HKUST-1 Metal-Organic Frameworks: Insights into Synthesizability and Hydrolytic Properties from Machine Learning Potentials. Rui Zheng; Xiaoyu Wu*; Jianwen Jiang*. *Under review at Journal of the American Chemical Society.*

Spatially Aware Deep Learning for Isotopologue Separation: Predicting 12CH₄/13CH₄ Selectivity in Metal-Organic Frameworks via Grid-Crystal Residual Networks. Chenrui Li†; Xiaoyu Wu*; Tianyu Guo; Linjiang Chen; Lin-Bing Sun*; Zhenghao Wu*; Lifeng Ding*. *Under review at Separation and Purification Technology.*

Peer-Reviewed Publications (29)

- Hang-Ou Qi†; **Xiaoyu Wu†**; Jiandong Pang; Guoliang Liu; Lifeng Ding*; Lin-Bing Sun*. "Pore Space Partition in an indolo[3,2-b]carbazole-based Zr-MOF via Post-Synthetic Linker Installation for Enhanced Selective Ethane/Ethylene Separation." *Inorganic Chemistry*, 2026 (Provisionally accepted)
- Hang-Ou Qi†; Xiao-Han Li†; Ze-Jiu Diao; Fan Li; **Xiaoyu Wu**; Lifeng Ding; Guoliang Liu*; Lin-Bing Sun*. "Efficient CO₂ Capture and Conversion from Flue Gas Enabled by an Indolo[3,2-b]carbazole-Based Copper Metal–Organic Framework." *ACS Materials Letters*, 2026 DOI
- Xiaoyu Wu**; Jianwen Jiang*. "Interrogating Synthetic Likelihood of Metal-Organic Frameworks: A Digital Discovery Perspective." *Chemical Science*, 2026 (Cover Article) DOI
- Xiaoyu Wu**; Quan Liu; Rui Zheng; Jianwen Jiang*. "Stability-aware, transferable discovery of metal-organic frameworks for selective adsorption of C₆ alkane isomers by machine learning." *Chemical Engineering Journal*, 2026 DOI
- Mingrui Zuo†; **Xiaoyu Wu†**; Manish Kumar Dinker*; Mengmeng Zhang; Zhenghao Wu; Liangdan Zhao; Chenrui Li; Tianyu Guo; Linjiang Chen; Lin-Bing Sun*; Lifeng Ding*. "Double-shell enhanced CO₂ capture in metal-organic cage-based porous liquids." *Separation and Purification Technology*, 2026 DOI
- Rui Zheng; Saad Aldin Mohamed; Quan Liu; **Xiaoyu Wu***; Jianwen Jiang*. "CO₂ adsorption-induced anisotropic mechanical response in an 8-fold interpenetrated diamondoid metal-organic framework." *Journal of Materials Chemistry A*, 2026 DOI
- Xiao Niu; Zhiming Zhang; **Xiaoyu Wu**; Yan Liu; Yong Cui*; Jianwen Jiang*. "Machine learning guided discovery of water stable metal-organic frameworks for photocatalytic hydrogen production." *Chemical Science*, 2026 (Cover Article) DOI
- Quan Liu*; Rui Zheng; Zhuolin Liang; Longlong Sun; **Xiaoyu Wu***; Gongping Liu. "Breathing pores orchestrated by multifunctional chemical sites in graphene-based conjugated microporous polymer membranes for efficient H₂ purification." *Fuel*, 2026 DOI
- Xiaoyu Wu**; Rui Zheng; Quan Liu; Jianwen Jiang*. "Digital Discovery of Synthesizable Metal-Organic Frameworks via Molecular Dynamics-Informed, High-Fidelity Deep Learning." *Advanced Functional Materials*, 2026 (Cover Article) DOI
- Guoliang Liu; Fan Li; Ze-Jiu Diao; Hua-Dong Li; **Xiaoyu Wu**; Hang-Ou Qi; Lifeng Ding; Lin-Bing Sun*. "Supramolecular framework crystallinity engineering via surface-confined polymerization for enhanced C₃H₆/C₂H₄ separation." *Journal of Materials Chemistry A*, 2025 DOI

11. Tianyu Guo; **Xiaoyu Wu***; Chenrui Li; Liangdan Zhao; Mingrui Zuo; Lin Liu; Lin-Bing Sun*; Lifeng Ding*. "From chemisorption to physisorption: A synergistic computational screening approach to exploit N-rich covalent organic frameworks for methyl iodide capture." *Chemical Engineering Journal*, 2025 DOI
12. **Xiaoyu Wu**; Quan Liu; Jianwen Jiang*. "Discovering robust metal-organic frameworks with open copper sites for precombustion CO₂ capture: Data-efficient exploration and exploitation by active learning." *Chemical Engineering Journal*, 2025 DOI
13. Quan Liu*; **Xiaoyu Wu**; Rui Zheng; Gongping Liu; Jianwen Jiang*. "Machine-learning-aided design of two-dimensional C₆₀ membranes for molecular-orientation-directed CO₂/C₂H₂ separation." *Journal of Membrane Science*, 2025 DOI
14. Yuanhong Yu; Xianyu Song*; Xu Yang; Chengjie Wang; **Xiaoyu Wu**; Yanglong Wang; Wenjun Xiang; Shuangliang Zhao; Honglai Liu. "Optimization of chitosan-based demulsifiers via interfacial displacement: A molecular dynamics and principal component analysis approach." *Separation and Purification Technology*, 2025 DOI
15. Quan Liu*; Zhuolin Liang; **Xiaoyu Wu**; Zhonglian Yang; Gongping Liu. "Graphene Oxide Membranes with Engineered Pocket Suction Water Environments for Selective C₁-C₂ Alcohol Dehydration." *ACS Sustainable Chemistry & Engineering*, 2025 DOI
16. Rui Zheng†; **Xiaoyu Wu†**; Saad Aldin Mohamed; Jianwen Jiang*. "Unraveling Guest-Induced Structural Transition in a Flexible Metal-Organic Framework for C₄ Alkane/Alkene Separation." *Chemistry of Materials*, 2025 DOI
17. Zhiming Zhang; Athulya Surendran Palakkal; **Xiaoyu Wu**; Jianwen Jiang*; Zhongyi Jiang*. "Discovering Ultra-Stable Metal-Organic Frameworks for CO₂ Capture from a Wet Flue Gas: Integrating Machine Learning and Molecular Simulation." *Environmental Science & Technology*, 2025 DOI
18. **Xiaoyu Wu**; Jianwen Jiang*. "Can large language models predict the hydrophobicity of metal-organic frameworks?." *Journal of Materials Chemistry A*, 2025 (Cover Article) DOI
19. **Xiaoyu Wu**; Xianyu Song; Yifei Yue; Rui Zheng; Jianwen Jiang*. "MOF-KAN: Kolmogorov-Arnold Networks for Digital Discovery of Metal-Organic Frameworks." *The Journal of Physical Chemistry Letters*, 2025 DOI
20. Liangdan Zhao; Chunyi Yu; **Xiaoyu Wu**; Mingrui Zuo; Qian Zhang; Qiuchen Dong*; Lifeng Ding*. "Computational Screening Guiding the Development of a Covalent-Organic Framework-Based Gas Sensor for Early Detection of Lithium-Ion Battery Electrolyte Leakage." *ACS Applied Materials & Interfaces*, 2025 DOI
21. **Xiaoyu Wu†**; Rui Zheng†; Jianwen Jiang*. "Leveraging Cross-Diversity Machine Learning to Unveil Metal-Organic Frameworks with Open Copper Sites for Biogas Upgrading." *Journal of Chemical Theory and Computation*, 2025 DOI
22. **Xiaoyu Wu**; Xianyu Song; Linjiang Chen*; Chunyi Yu; Liangdan Zhao; Mingrui Zuo; Chenrui Li; Heechae Choi; Jianwen Jiang*; Lifeng Ding*. "Accurate and interpretable machine learning with Pore+ descriptors for iodide capture in metal-organic frameworks." *Separation and Purification Technology*, 2024 DOI
23. **Xiaoyu Wu**; Jianwen Jiang*. "Precision-engineered metal-organic frameworks: fine-tuning reverse topological structure prediction and design." *Chemical Science*, 2024 (Cover Article) DOI
24. Chengjie Wang; Pingli Liu; Jinzhou Zhao; Juan Du; Xiang Chen; **Xiaoyu Wu***; Xianyu Song*; Guan Wang; Jinming Liu; Yonghua Xiao*. "Topological isomerism design of demulsifiers: A fusion of molecular dynamics simulation and experiment." *Journal of Molecular Liquids*, 2024 DOI
25. Yunsheng Ma; Xiaoyan Tang; Ming Chen; Akio Mishima; Liangchun Li; Akihiro Hori; **Xiaoyu Wu**; Lifeng Ding; Shinpei Kusaka; Ryotaro Matsuda. "Design of a MOF based on octa-nuclear zinc clusters realizing both thermal stability and structural flexibility." *Chemical Communications*, 2022 DOI
26. **Xiaoyu Wu**; Yu Che; Linjiang Chen*; Eric Jean Amigues; Ruiyao Wang; Jinghui He; Huilong Dong; Lifeng Ding*. "Mapping the Porous and Chemical Structure-Function Relationships of Trace CH₃I Capture by Metal-Organic Frameworks using Machine Learning." *ACS Applied Materials & Interfaces*, 2022 DOI
27. **Xiaoyu Wu**; Linjiang Chen*; Eric Jean Amigues; Ruiyao Wang; Zhongfu Pang; Lifeng Ding*. "In Silico Tuning of the Pore Surface Functionality in Al-MOFs for Trace CH₃I Capture." *ACS Omega*, 2021 DOI
28. Meng-Yao Chao; Jing Chen; Xiao-Yu Wu; Rui-Yao Wang; Pei-Pei Wang; Lifeng Ding*; David J. Young; Wen-Hua Zhang*. "Unconventional Pyridyl Ligand Inclusion within a Flexible Metal-Organic Framework Bearing an N,N'-bis-(2-diethylformamide (DEF)-Solvated Cd₅ Cluster Secondary Building Unit." *ChemPlusChem*, 2020 DOI
29. Yuxin Ge; Hao Zhou; Yujin Ji; Lifeng Ding*; Yuanyuan Cheng; Ruiyao Wang; Siyuan Yang; Yufeng Liu; **Xiaoyu Wu**; Youyong Li*. "Understanding Water Adsorption and the Impact on CO₂ Capture in Chemically Stable Covalent Organic Frameworks." *The Journal of Physical Chemistry C*, 2018 DOI

Teaching and Mentorship

- Teaching assistant and co-supervisor across Introduction to Chemistry; Introductory Inorganic, Physical, and Organic Chemistry I-II; Laboratory Techniques; Preparative Chemistry; and Analytical Chemistry.
- Mentored MSc, BSc, and PhD researchers on MOF/COF simulation, adsorption and separation, framework mechanics and stability, machine learning, and LLM-assisted literature mining.
- Mentees include Rui Zheng (current PhD student, NUS), Tianyu Guo (now PhD researcher, HKUST), Mingyue Ma, Yixu Cheng, and Daeun Jung.

Selected Conference Presentations

- AIChE Annual Meeting, Boston, United States (2025) - oral: Rejuvenating digital discoveries of MOFs through screening, featurization, and diversity-driven machine learning.
- AIChE Annual Meeting, Boston, United States (2025) - oral: Guest-induced structural transition in a flexible MOF for C₄ alkane/alkene separation.
- AIChE Annual Meeting, Boston, United States (2025) - poster: Large language models for MOF hydrophobicity prediction.

- MOF 2024, Singapore (2024) - poster: Construction of hypothetical MOFs driven by reticular design.
- AIChE Annual Meeting, San Diego, United States (2024) - poster: Cross-database discovery of open-Cu(II) MOFs for biogas upgrading.

Research Funding and Computational Resources

- Participant, LCER Phase 2 programme (SGD 23,481,120).
- Participant, LCER Phase 1 projects LCERFI01-0015 U2102d2004 and LCERFI01-0033 U2102d2006 (SGD 7,534,320).
- Participant, NRF-CRP26-2021RS-0002 (SGD 8,871,670).
- Awarded large-scale NSCC and NUS CRP computational allocations, including multi-million CPU-hour and substantial GPU-hour projects.

Peer Review

Reviewer for Chemical Engineering Journal, Water Research, Journal of Colloid and Interface Science, Computational Materials Science, Journal of Chemical Information and Modeling, Journal of Physical Chemistry B/C, ACS Applied Materials & Interfaces, Energy Conversion and Management, and related journals.

Selected Research Outputs

- Precision-Engineered MOF database and reverse-topological design workflow (94,823 structures).
- Open-copper-site MOF databases and cross-diversity machine-learning models.
- MD-informed deep-learning dataset for synthesizable MOF discovery.
- Active-learning dataset for precombustion CO₂ capture; Pore+ descriptors; MOF-KAN; fine-tuned LLMs for MOF hydrophobicity.

Professional Membership

Member of the Royal Society of Chemistry (MRSC).